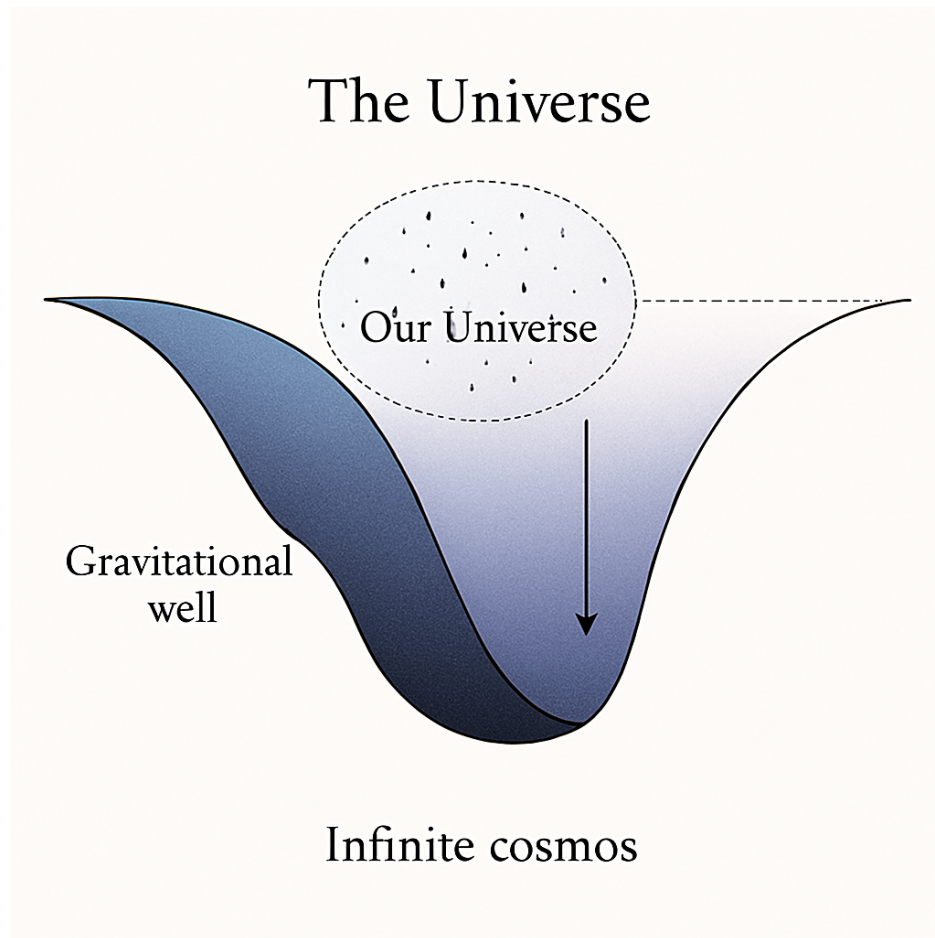


Toward a Gravitational Framework for Cosmology

Initial Mathematical Scaffolding for Tom Young's Theory

May 14, 2025



Our observable universe, falling along the slope of a vast gravitational well in an infinite cosmos.

Introduction: Expanding Without Expansion

The goal of this document is to propose a minimal, mathematically consistent foundation for Tom Young’s gravitational-well theory of cosmic redshift and expansion — a theory that eliminates the need for dark energy, inflation, or a singular Big Bang.

This framework replaces the assumption of metric expansion with the notion that light is redshifted as it climbs out of a large-scale gravitational potential — sourced by mass and structure beyond our observable universe.

1. The Geometric Setup

We begin by describing a simple geometry:

Let us define:

- $\Phi(r)$ — the gravitational potential at radial coordinate r from a large-scale external structure.
- $r = 0$ — our current location within OU (Our Universe).
- $r = R_{\text{OUEH}}$ — our observable boundary (the OUEH).

We assume OU is embedded in a larger TU, and that $\Phi(r)$ varies slowly but significantly across our observable horizon.

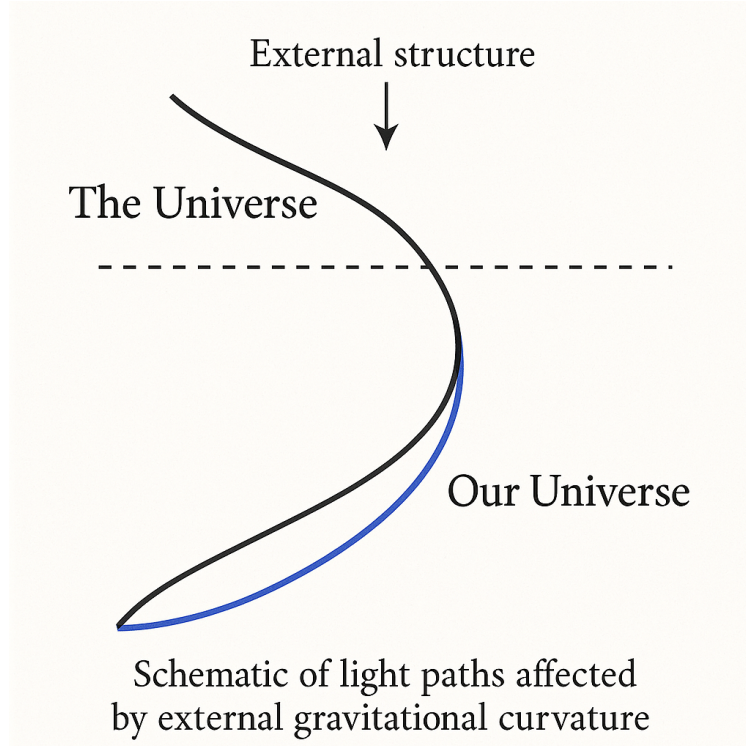


Figure 1: Schematic of photon paths affected by external gravitational curvature.

2. Gravitational Redshift

The key assumption: **cosmic redshift arises not from expanding space, but from gravitational redshift across $\Phi(r)$.**

The gravitational redshift between emission at radius r and reception at r_0 is approximately:

$$z = \frac{\Delta\Phi}{c^2} = \frac{\Phi(r_0) - \Phi(r)}{c^2}$$

Assuming that $\Phi(r)$ decreases as we look outward (deeper into the well), this gives redshift naturally — without invoking scale factor $a(t)$ or Hubble flow.

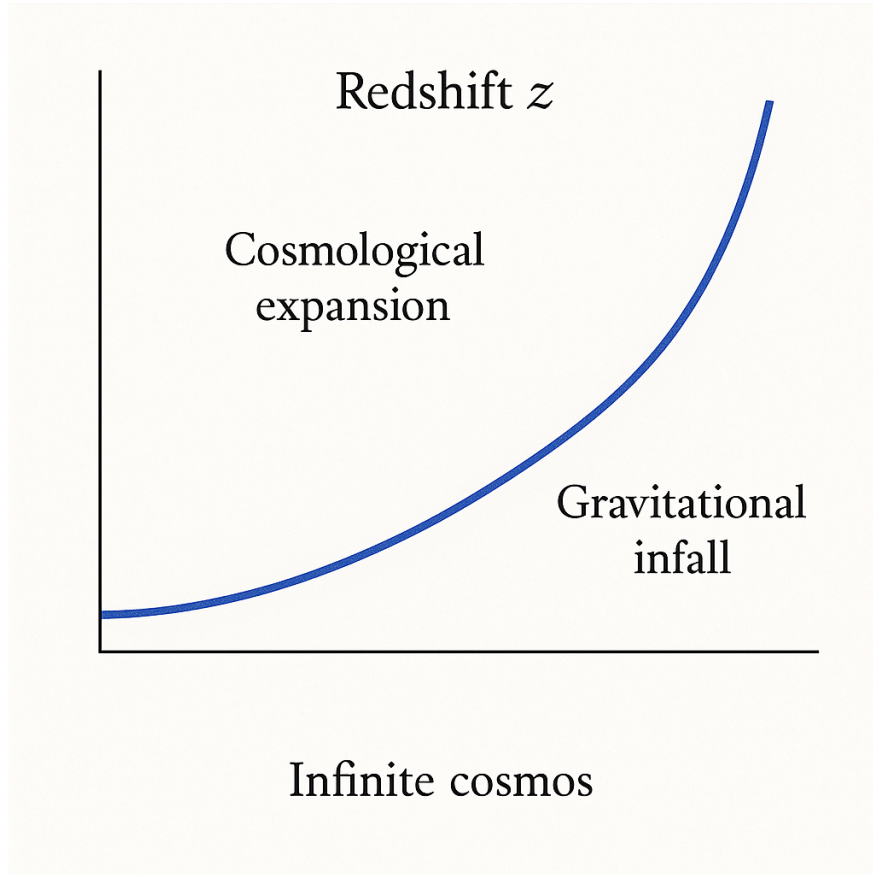


Figure 2: Comparison of redshift predictions from standard expansion vs. gravitational infall.

3. Light Propagation and Distance Measures

Distance measures (luminosity distance D_L , angular diameter distance D_A) can be derived using null geodesics in the assumed metric:

$$ds^2 = -(1 + 2\Phi(r))c^2dt^2 + (1 - 2\Phi(r))dr^2 + r^2d\Omega^2$$

This simplified Schwarzschild-like metric allows us to compute photon paths and thus:

$$D_L(z) = (1 + z)^2 D_A(z)$$

Which can be fitted against observed SN Ia data.

4. Mapping Redshift to CMB Fluctuations

Tom's proposed observational test — that redshift anisotropies and CMB fluctuations align — suggests:

$$\delta z(\hat{n}) \sim \delta T_{\text{CMB}}(\hat{n})$$

Where both quantities vary over direction $\hat{n}$ on the sky. This could be tested with Planck and Pantheon+ datasets.

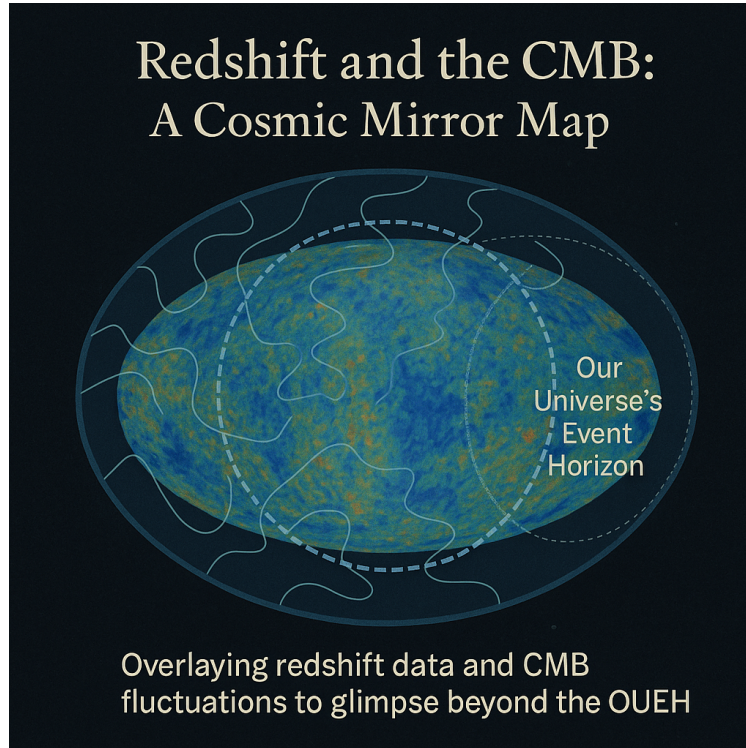


Figure 3: Overlaying CMB and redshift data to reveal external gravitational structures.

5. Further Directions: A Roadmap for Collaboration

This framework is not complete. It is a scaffolding for future thinkers to build upon. Possible extensions include:

- Numerical modeling of $\Phi(r)$ from synthetic TU mass distributions.
- Bayesian fitting of gravitational-redshift models to SN Ia and BAO data.
- Simulating large-scale structure growth without dark matter.
- Formal exploration of gravitational wave propagation in TU.

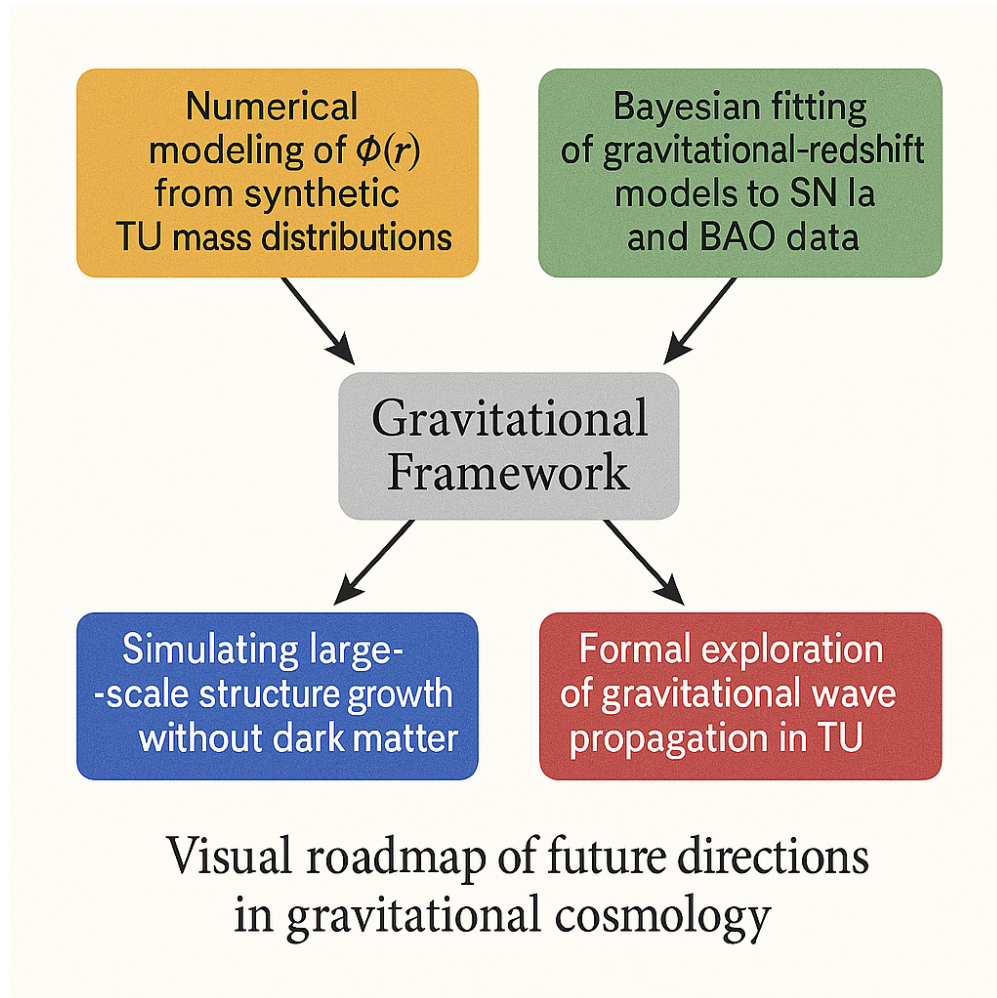


Figure 4: Visual roadmap of future directions in gravitational cosmology.

Conclusion

This minimal model demonstrates that Tom's theory can be formulated mathematically in a consistent way. While much work remains, nothing in the known laws of physics rules out a redshift driven by gravitational curvature.

The goal now is to refine, test, simulate — and map the invisible structure of The Universe.